Nicol Emanuele

MSc Computer Science Student | Software Engineering, Algorithms & Complex Systems

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Skills

Software Engineering & Systems:
Rust, **Python** (FastAPI), backend development, algorithms and data structures, numerical computation, performance-oriented design, correctness-critical logic, distributed systems fundamentals.

Algorithms, Complexity & Optimization:
Familiarity with algorithmic complexity analysis, asymptotic reasoning, and trade-offs between time and space complexity applied to practical system design and optimization problems.

Data Analysis & Reliability:
Data analysis pipelines, time-series analysis, feature engineering, anomaly detection, statistical evaluation, CI/CD pipelines, GitHub Actions, GitLab CI, Docker, AWS (EC2, VPC), testing, debugging, automation, technical documentation.

Selected Projects

- **Telemetry Analysis & Anomaly Detection (Python, Time-Series)**
  Designed a modular system for analyzing multivariate spacecraft telemetry (NASA SMAP and MSL datasets), focusing on data reliability, feature extraction, and scalable analytical pipelines for anomaly detection.
- **Algorithmic Analysis of Structured Event Data (Python)**
  Conducted an end-to-end analytical pipeline on structured gameplay data, covering data cleaning, exploratory analysis, feature engineering, statistical evaluation, and outcome prediction. All steps were documented in reproducible Jupyter notebooks.
- **System Performance Monitoring Tool (TypeScript)**
  Designed and implemented a real-time system monitoring application to track CPU, memory, and runtime metrics, with a focus on data collection, visualization, and reliability of long-running processes.

Work Experience

Provably, Warsaw Feb 2025 – Present

Software Engineering Intern

- Implemented **algorithmic logic** inside constraint-based systems, translating high-level specifications into **low-level arithmetic representations** with strict **correctness requirements**.
- Worked on **performance- and correctness-critical components**, reasoning about **numerical stability**, **constraint efficiency**, and **system-level trade-offs**.
- Contributed to the **full development workflow**, from prototyping algorithmic components to **automated testing** and **CI pipelines** using **GitHub Actions**.

Run Times Machines, Zurich Aug 2023 – Dec 2023

Software Engineering Intern

- Implemented **verification algorithms** operating on **explicit binary representations**, managing **bit-level logic** and **correctness constraints** within **security-critical systems**.
- Designed and maintained **modular verification components**, balancing **low-level binary manipulation** with **clean, maintainable system architecture**.

Education

MSc Computer Science 2023 – Present
University of Milano-Bicocca (Erasmus: Stockholm University)
Focus on **algorithms**, **optimization**, **software engineering**, **distributed systems**, **and reliable system design**.